LAS Honors Project Proposal Form

To be completed one term prior to graduation (or presentation at Honors poster session)

Student Information

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Minor(s): Computer Science
Academic Advisor(s): Allyn Kolar

Project Information

Title of Project: Towards Coloc-Confirmed TWAS of Ankylosing Spondylitis to Prioritize Tissue-Specific Drug Targets
Type of Project: Research
Project Advisor and Department: (Advisor TBD) | Department of Computer Science

Review Process

Proposals will be reviewed by a committee of faculty in LAS and must be approved before projects are started. The Committee will ensure that a project can be completed on time, that it makes a demonstrable contribution to scholarship, and that it will be done well. Since the reviewers may not be in your area of study, please define all technical terms and abbreviations and write in a manner so that your ideas will be understood by others outside of your major.

Project Proposal

The purpose of the LAS Honors project is to advance knowledge through scholarship. Scholarship includes a range of activities such as research, writing, and original composition/performance. An LAS Honors project can be **research** (using an organized and logical process of inquiry to answer questions) or **creative** (crafting an artistic creation).

1. For **research** projects, what research question(s) do you plan to address, or which hypothesis do you plan to test in this project? For **creative** projects, what will your artistic or societal contribution be?

- RQ1: Which genes show significant transcriptome-wide association with ankylosing spondylitis (AS) in immune-relevant tissues?
- RQ2: For those genes, do the AS GWAS signal and cis-eQTL signal share a single causal variant (Bayesian colocalization)?
- Hypotheses. (H1) Immune-relevant tissues (whole blood, spleen, small intestine/terminal ileum) will be enriched for significant TWAS hits. (H2) A subset of TWAS hits will reach strong colocalization (PP4 ≥ 0.7), increasing confidence that gene expression is on the causal path to AS.

2. Provide a short description of your project. (Approximately 75-100 words)

This project integrates ankylosing spondylitis (AS) genome-wide association study (GWAS) summary statistics with GTEx v8 expression quantitative trait locus (eQTL) models to find gene-level, tissue-aware associations using S-PrediXcan, a transcriptome-wide association study (TWAS). I will then apply

Bayesian colocalization (COLOC) to test whether disease and expression signals share a causal variant, filtering linkage disequilibrium (LD) bystanders. The approach combines supervised machine learning (gene-expression prediction) with Bayesian inference for causal prioritization. Outputs include a ranked list of coloc-confirmed genes with tissues, effect directions, and pathway/druggability notes, a fully reproducible pipeline, an ISU poster, and a short manuscript.

3. How will your project be different from previous work in this area? If your project is part of a larger group project or lab, clearly describe your unique contribution to the project and how your project fits into the larger project. Clarify what your group is doing ("we") and what you will do ("I").

   Prior AS genetics primarily reports SNP-level hits and pathway enrichments. I will move beyond SNPs to gene-level signals and require statistical colocalization, a stricter causal criterion that many prior papers omit. I will: (i) harmonize public AS GWAS summary statistics, (ii) run multi-tissue TWAS with rigorous multiple-testing control, (iii) perform COLOC with sensitivity analyses, and (iv) release an open, end-to-end pipeline. If the lab group explores related loci or methods, they may provide general guidance/resources, while I will own data harmonization, analysis, figure generation, biological annotation, and manuscript drafting.

4. Bibliography: This preliminary bibliography should include the sources that you reviewed as you prepared your project proposal. A more complete bibliography will be expected with your finished project. Cite all sources in a style appropriate for your discipline. Provide full bibliographic entries not just web links. (List 4-6 sources)

- Gamazon, E. R., et al. (2015). A gene-based association method for mapping traits using reference transcriptome data. Nature Genetics, 47(9), 1091–1098. https://doi.org/10.1038/ng.3367

- Barbeira, A. N., et al. (2018). Exploring the phenotypic consequences of tissue-specific gene expression variation inferred from GWAS summary statistics. Nature Communications, 9, 1825. https://doi.org/10.1038/s41467-018-03621-1

- Giambartolomei, C., et al. (2014). Bayesian test for colocalisation between pairs of genetic association studies. PLoS Genetics, 10(5), e1004383. https://doi.org/10.1371/journal.pgen.1004383

- Wang, G., Sarkar, A., Carbonetto, P., & Stephens, M. (2020). A simple new approach to variable selection in regression, with application to genetic fine-mapping (suSiE). Annals of Applied Statistics, 14(2), 985–1013. https://doi.org/10.1214/19-AOAS1289

- Benner, C., et al. (2016). FINEMAP: Efficient variable selection using summary data from genome-wide association studies. Bioinformatics, 32(10), 1493–1501. https://doi.org/10.1093/bioinformatics/btw018

- GTEx Consortium. (2020). The GTEx Consortium atlas of genetic regulatory effects across human tissues. Science, 369(6509), 1318–1330. https://doi.org/10.1126/science.aaz1776

5. List any coursework (include course titles) or experiences that have prepared you for your work on this project.

   Courses: Intro to Algorithms; Probability & Statistics; Statistical Modeling; Machine Learning; Data Management/SQL

   Experience: Data Science & Analytics Intern, John Deere ISG (Summer 2025)—built/automated data pipelines, version control, and reproducible workflows; extensive Python (pandas, scikit-learn), R, and Git/GitHub use.

6. Describe the procedures and methods you will use to complete your project. If you will be carrying out scientific research, explain the specific types of tests or analyses, being sure to avoid jargon.

   I will download public, de-identified AS GWAS summary statistics and GTEx v8 eQTL models. After aligning genome builds/alleles, I will run S-PrediXcan to perform a TWAS, which uses supervised ML models (trained on GTEx) to predict gene expression from genetic variants and test its association with AS. For TWAS-significant genes, I will run Bayesian colocalization (COLOC) in ±500 kb windows to estimate the probability of a shared causal variant (PP4, the posterior probability). Genes with PP4 ≥ 0.7 will be prioritized and annotated for pathways/druggability. (Time-permitting) I will fine-map top loci (suSiE/FINEMAP). All code, environment files, and figure scripts will be released for full reproducibility.

7. For creative/innovative projects only. In addition to creating an original end product, the student should place their work within its proper context in its field or craft and reflect on the vision, process, and goals that shaped this creative experience. Please discuss your plans for the critical analysis of your final project.

   NA

8. Outline your proposed project schedule. Provide a timeline of anticipated dates to complete major stages of your project.

   Proposed schedule (Spring 2026):

- Prep (Nov–Dec 2025): Confirm mentor; set up repo/env; acquire datasets; dry-run on one locus.
- Jan 13–Feb 7: Run multi-tissue TWAS; QC; initial figures.
- Feb 10–Mar 7: COLOC for all TWAS hits; sensitivity analyses; figure drafts.
- Mar 10–Mar 28: Biological/pathway annotation; prioritized gene list; internal review.
- Mar 31–Apr 18: (Optional) fine-mapping; finalize figures & robustness checks.
- Apr 21–May 2: Write/polish manuscript & poster; submit to venue; present at ISU event.

9. In addition to the Honors Poster presentation, how else will you share the results of your project?

- Publication: Submit a short manuscript to ML4H or CHIL (or a suitable genetics/biomed short-format venue).

- Public GitHub repository with code, environment, and dataset/model cards.

10. You can earn 1-3 academic credits for your project based on the amount of work for your project. Please consult with your project advisor and register for the appropriate number of credits.

    Number of Credits: ~2
    Course number: HON 4900 & DS 401
    (you will register for in term of Honors poster presentation)

11. Will you be completing your project with another student from another academic discipline?

    Yes☐ No☒

    If yes, what is the student's name and their degree program? Click here to enter text.

Responsible Research/IRB Approval

Honors capstone projects that involve human participants require review if results will be disseminated beyond the project advisor(s), Honors Program poster presentations, Undergraduate Research and Create Expression seminars, or at the State Capitol. Presentations given in the latter setting are considered a continuation of the education process or professional training (e.g., experience in public speaking). For more information regarding IRB approval please visit https://www.compliance.iastate.edu/committees/irb

Please select one option:
☒ This project does not require outside approval or training

☐ This project is covered by approval already granted for a larger project

☐ Application for approval has been submitted to the Office for Responsible Research. Date submitted: Click here to enter text.

Signature Page

☒ I have read the Project Proposal Instructions and agree to the requirements for the completion of this project.

Bhargav Yellepeddi **11/22/2025**

Student Signature Date

☑ I have read the student's proposal and the Guidelines for Project Advisors and am willing to mentor the student in the completion of this project.

Ratul Chowdhury 01/22/2026

Project Advisor Signature Date

☒ I am aware of my advisee's intention to complete an Honor Project as described in the proposal.

Allyn Kolar 1/22/2026

Academic Advisor Signature Date

Email paperwork to lashnrs@iastate.edu by September 21 for Fall submission or February 21 for Spring submission.